

Team Report-Outs, Synthesis, and What's Next

Module 3, Part 3 (October 17)

QBA 5330

Quick Reminder: The Worksheet

1. Random assignment
2. Confounders (a **new** one)
3. Bias in measurement
4. Statistical vs. practical significance
5. A **specific** stronger design
6. Your team's verdict

Team Report-Outs (30 minutes)

Each team: ~3–4 minutes on your assigned item.

We'll track agreement and disagreement as we go.

A Representative Synthesis

- **Strongest confounders:** team quality/resources at time of assignment; reverse causation; self-selection into management
- **Most concrete stronger design:** randomly assign manager-behavior coaching to a subset of managers, hold a comparable group back as a control, compare outcomes afterward
- **Class verdict:** suggestive and internally consistent, but still observational, a randomized pilot would meaningfully strengthen the claim

Where This Fits

We just did:

- **Quantify uncertainty** (evaluating the evidence)
- A first pass at **recommend an action** (proposing a stronger design)

Using real, messy business evidence, not a clean textbook dataset.

Module 3 Wrap-Up

What Module 3 added to your toolkit:

- One-sample and two-sample comparisons
- Paired designs
- ANOVA and Tukey's HSD
- The judgment to ask: does this comparison support a **causal** claim?

Looking Ahead to Module 4 (11/13)

From comparing groups to modeling relationships.

Regression: the start of *build a model*, the next step of the analytics decision cycle.

From “is there a difference?” to “how does one thing relate to another, and what can we predict?”